YEAR 11 PHYSICS, AQA GCSE, PART 2 OF 4

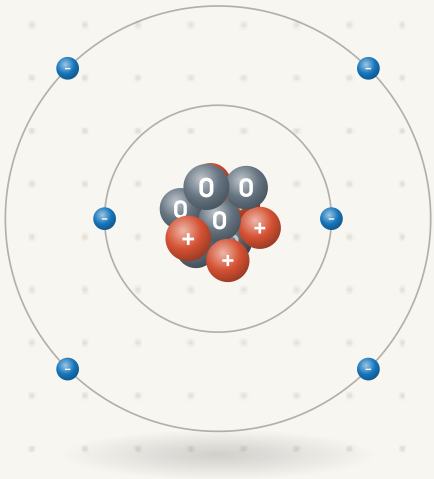
Atoms & Nuclear Radiation.

Part 2 of 4 Lessons 3–4

IN THIS BOOKLET

- Lesson 3: From plum pudding to the nucleus
- Lesson 4: Energy levels and the nucleus

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

How this booklet works.

This is the booklet for **Lessons 3–4** of the Atoms and Nuclear Radiation unit. It builds on the lessons before it; the "Do now" questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
Explanations and diagrams. Fill in the slide examples with your teacher, then try "Your turn" alone.

TRY IT
Practice that builds up, then a challenge. Quick checks from the slides follow some lessons.

LAB
Record what you find in a simulation or a model experiment.

EXAM CORNER
AQA-style questions with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an "I can" check to colour in.

! **Revising at home?** (1) Read the Learn it pages. (2) Cover them and answer the "Cover the page" questions in Revise it. (3) Check with the quick answers at the back. (4) Go through the matching slides in the teaching deck, and try the simulations.

The route through the unit (this booklet highlighted)

01 Inside the atom

02 Isotopes and ions

03 From plum pudding to the nucleus
Slides 9–17

04 Energy levels and the nucleus
Slides 18–25

05 Radioactive decay

06 Nuclear equations

07 Properties and uses of radiation

08 Half-life and random decay

09 Half-life calculations

10 Contamination and irradiation

Teaching deck: **panphy.app/gcsephy/decks/atoms-and-radiation**. Simulations: **panphy.app/simulations/atomic_models.html** and **panphy.app/simulations/nuclear_decay.html**

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LESSON

03 From plum pudding to the nucleus.

AQA Physics 4.4.1.3; Combined Science 6.4.1.3

BIG QUESTION

Nobody has ever seen inside an atom. So how did scientists find out that almost all of its mass is squeezed into a tiny nucleus?

BY THE END YOU CAN

- describe the plum pudding model and why it was proposed
- describe the alpha particle scattering experiment
- explain how each result led to the nuclear model
- compare the plum pudding and nuclear models

model | plum pudding model | alpha particle | alpha scattering | deflected | nuclear model | evidence

SLIDES AND SIMULATIONS

Teaching deck, slides 9–17

Timeline of models, the plum pudding model, alpha scattering and the nuclear model.

panphy.app/gcsephy/decks/atoms-and-radiation/#9

Atomic Models simulation

panphy.app/simulations/atomic_models.html

RETRIEVE

Do now

From earlier lessons. No notes!

1 What particles are in the nucleus?

Protons and neutrons.

2 What is the overall charge of a nucleus? Why?

Positive: protons are positive and neutrons have no charge.

3 The radius of a nucleus is less than what fraction of the atom's radius?

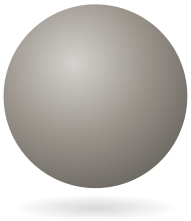
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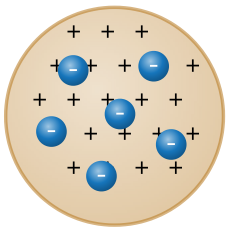
EXPLAIN

How the model changed

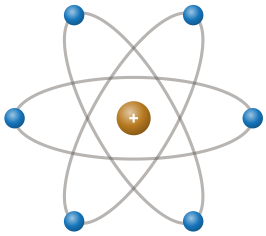
A model that kept changing



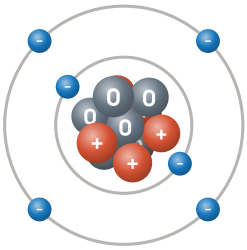
Before 1897
Tiny solid spheres that cannot be divided



1904
Plum pudding: after the electron was discovered



1911
Nuclear model: from alpha scattering



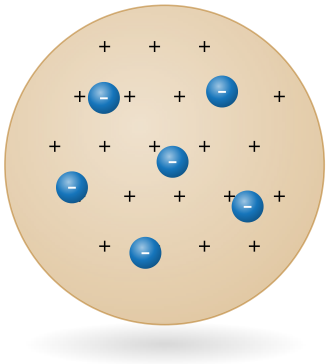
1913
Bohr model: electrons in energy levels

Scientists use **models** to explain things too small to see. When new **evidence** does not fit a model, the model is changed or replaced.

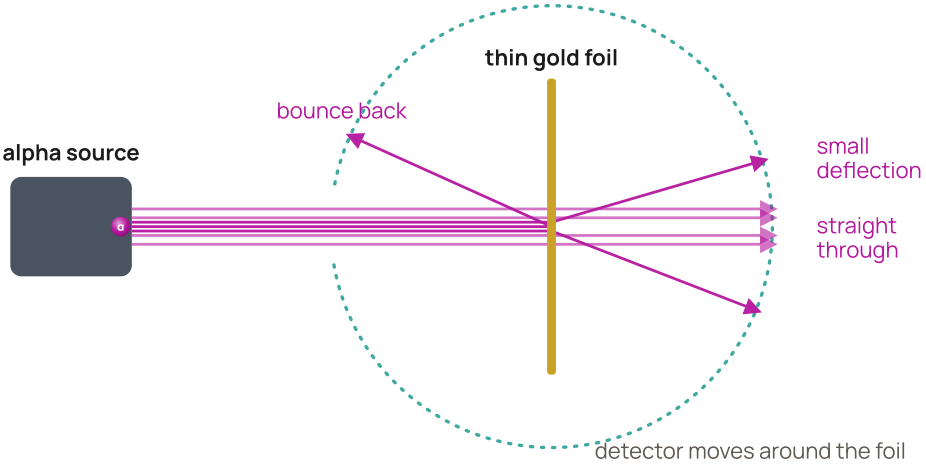
The plum pudding model (1904)

In 1897 J. J. Thomson discovered the **electron**, a tiny negative particle that comes from inside atoms. So atoms *can* be divided into smaller parts.

This led to the **plum pudding model**: the atom is a **ball of positive charge** with **negative electrons embedded in it**. The positive charge and the mass are spread through the whole atom. There is **no nucleus**.



The alpha scattering experiment (1909)



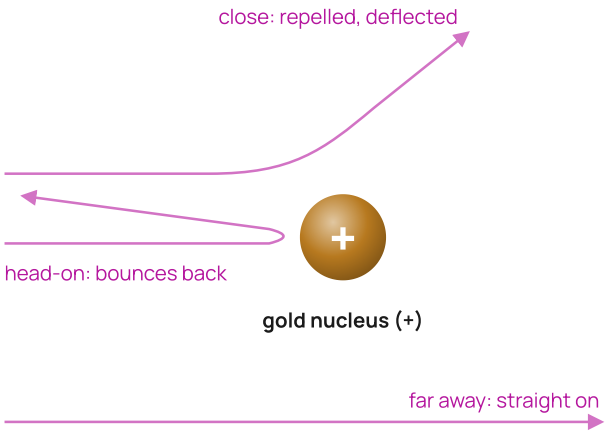
Geiger and Marsden, working with Rutherford, fired alpha particles at a very thin gold foil and counted them at every angle.

Alpha particles are small, fast and positively charged. The plum pudding model predicted that they would all pass **straight through** the foil, with at most tiny deflections, because the

positive charge was spread out too thinly to push them far off course.

What they saw, and what it means

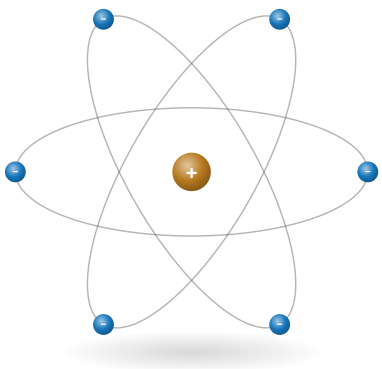
Observation	Conclusion
Most alpha particles went straight through.	The atom is mostly **empty space**.
Some were deflected through large angles.	The centre of the atom is **positively charged**: it repels the positive alpha particles.
A very few bounced back.	The **mass** and charge are concentrated in a **tiny centre**: the nucleus.



Close-up of one gold nucleus: not to scale.

The nuclear model (Rutherford, 1911)

	Plum pudding	Nuclear model
Positive charge	spread through the atom	in a tiny nucleus
Mass	spread through the atom	concentrated in the nucleus
Empty space?	no	mostly empty space
Electrons	embedded in the positive ball	outside the nucleus



! The plum pudding model could not explain alpha particles bouncing back. The new evidence did not fit, so the model was **replaced** by the nuclear model.

PRACTISE

Try it

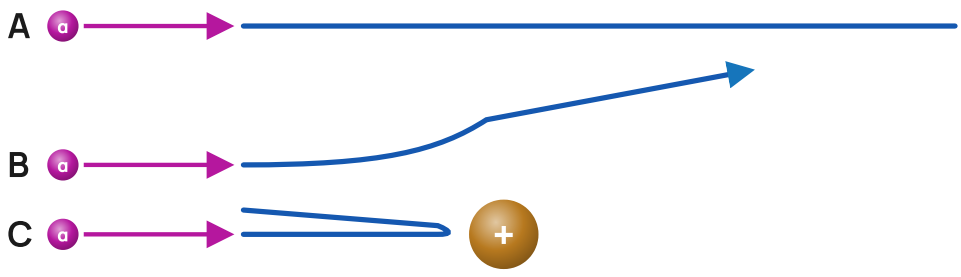
1 Put the models of the atom in the order they were proposed. Write 1 (first) to 4 (latest) in the boxes.

3	A tiny, positive nucleus with electrons outside it
1	Tiny solid spheres that cannot be divided
4	Electrons orbit the nucleus at fixed distances (energy levels)
2	A ball of positive charge with electrons embedded in it

2 Draw a line from each observation to the conclusion it led to. (The answer edition shows the matching letter.)

1 Most alpha particles passed straight through.	→ B	A The mass is concentrated in a tiny centre.
2 Some alpha particles were deflected.	→ C	B Most of the atom is empty space.
3 A very few alpha particles bounced back.	→ A	C The centre of the atom has a positive charge.

3 Three alpha particles, A, B and C, move towards a gold nucleus. Continue the path of each one.



gold nucleus

A passes far from the nucleus, B passes close to it and C is heading straight for it.

A carries straight on. B is repelled and deflected away from the nucleus. C slows, stops and bounces back the way it came.

4 Explain why the plum pudding model cannot explain alpha particles bouncing back.

In the plum pudding model the positive charge and mass are spread out through the atom. That is not concentrated enough to repel a fast, positive alpha particle strongly enough to turn it back. Bouncing back needs a tiny, dense, positive centre.

5 Is Mia right? Explain your answer.

> **Mia:** Most alpha particles went straight through, so the experiment proved the plum pudding model was right.

No. The plum pudding model also predicts that most go straight through, so that result does not decide between the models. The large deflections and the bounce-backs did not fit the plum pudding model, so it was replaced.

6 CHALLENGE The gold foil was only a few hundred atoms thick. Suggest why a very thin foil was needed.

Alpha particles are easily stopped, so they need thin foil to pass through at all. Each one then meets only a few nuclei, so each large deflection comes from one close approach to a single nucleus.

AQA-STYLE

Exam corner

Write in full sentences. Show every step of a calculation.

Get the marks: In a 'compare' question, say what each model says about the same feature: charge, mass, empty space, electrons.

7 Describe the differences between the plum pudding model and the nuclear model of the atom. [4 marks]

Any four: plum pudding has the positive charge spread out; nuclear model has it in a tiny nucleus (1). Mass spread out vs concentrated in the nucleus (1). Plum pudding has no empty space; the nuclear atom is mostly empty space (1). Electrons embedded in the positive charge vs electrons outside the nucleus (1).

8 In the alpha scattering experiment, most alpha particles passed straight through the gold foil, but a very small number were deflected by more than 90°. Explain what each result showed about the structure of the atom. **[4 marks]**

Most passing straight through shows the atom is mostly empty space (2). A very small number deflected by more than 90° shows there is a tiny centre (1) where the positive charge and mass are concentrated, which repels alpha particles (1).

REVIEW LATER

Revise it

Use this box after the lesson and before tests.

SUMMARY

- Models change when new evidence does not fit them.
- Plum pudding (1904): a ball of positive charge with electrons embedded; no nucleus.
- Alpha scattering: most straight through (empty space), some deflected (positive centre), a very few bounced back (mass in a tiny centre).
- Nuclear model (1911): tiny, dense, positive nucleus with electrons outside.
- The nuclear model replaced the plum pudding model.

COVER THE PAGE AND ANSWER

1 What discovery led to the plum pudding model?

2 What were alpha particles fired at?

3 What did bounce-backs show?

4 Where is the positive charge in the nuclear model?

I CAN…

- describe the plum pudding model
- describe the alpha scattering experiment
- link each result to a conclusion about the atom
- compare the plum pudding and nuclear models

LESSON

04 Energy levels and the nucleus.

AQA Physics 4.4.1.1–4.4.1.3; Combined Science 6.4.1.1–6.4.1.3

BIG QUESTION

Neon signs glow red and sodium street lamps glow orange. Each element gives out its own colours of light. What are its electrons doing?

BY THE END YOU CAN

- describe Bohr's model of electrons in energy levels
- explain how electrons change level when they absorb or emit electromagnetic radiation
- describe how protons and then neutrons were discovered
- explain why scientific models change

Bohr model · energy level · absorb · emit · electromagnetic radiation · proton · neutron · peer review

SLIDES AND SIMULATIONS

Teaching deck, slides 18–25

The Bohr model, energy levels, protons then neutrons, and why models change.

panphy.app/gcsephy/decks/atoms-and-radiation/#18

Atomic Models simulation

panphy.app/simulations/atomic_models.html

RETRIEVE

Do now

From earlier lessons. No notes!

1 Which experiment showed that atoms have a tiny nucleus?

The alpha scattering experiment.

2 What is the charge on an alpha particle?

Positive (+2).

3 In the nuclear model, where are the electrons?

Outside the nucleus.

EXPLAIN

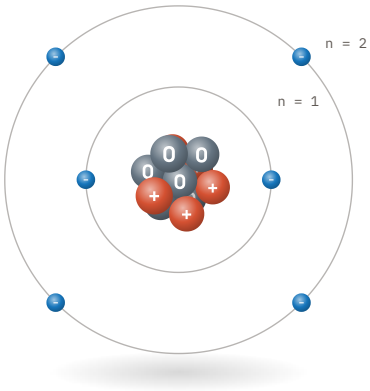
Filling in the details

Bohr's model (1913)

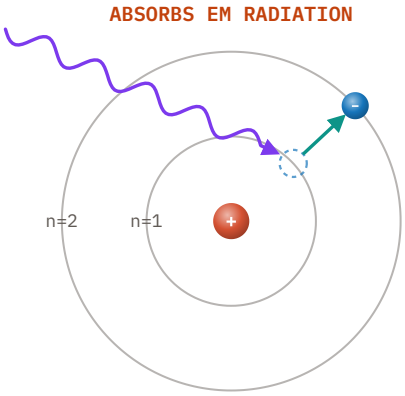
Niels Bohr adapted the nuclear model. He suggested that electrons orbit the nucleus **at specific distances**. Each allowed distance is an **energy level**.

Bohr's theoretical calculations **agreed with experimental observations** of the light given out by atoms. That agreement is why other scientists accepted his model.

A carbon atom has 2 electrons in the first energy level (n = 1) and 4 in the second (n = 2).

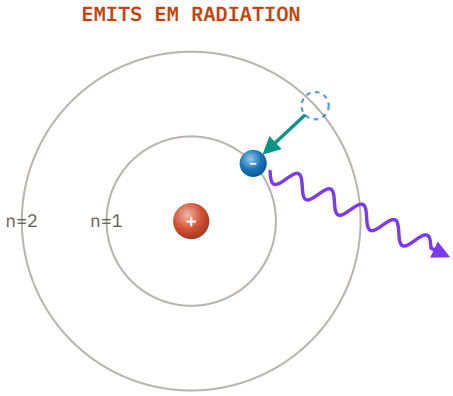


Electrons can change energy level



moves further out

higher energy level



moves closer in

lower energy level

The electron…	Moves…	Energy level
absorbs electromagnetic (EM) radiation	**further from** the nucleus	higher
emits electromagnetic (EM) radiation	**closer to** the nucleus	lower

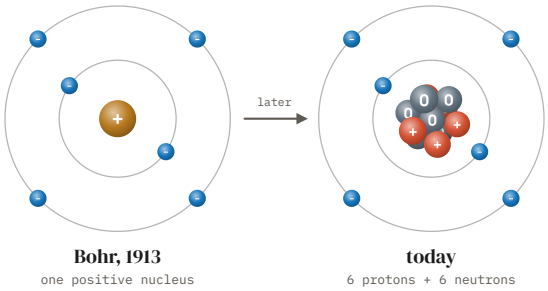
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THINK OF A LADDER

An electron can stand on a rung, but not between rungs. To climb up it must take in exactly the right amount of energy; when it drops down, it gives that energy out as EM radiation. This is why each element gives out its own colours of light.

Where the model breaks: the energy levels are not evenly spaced like rungs, and electrons are not little balls standing on them.

Filling in the nucleus



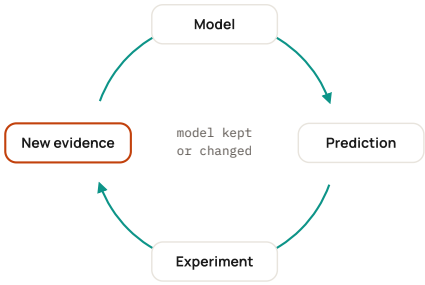
Bohr pictured one positive nucleus. Today we draw its protons and neutrons.

Later experiments (1919) showed that the positive charge of a nucleus is made of smaller particles, each with the same positive charge: **protons**.

In 1932 **James Chadwick** gave the experimental evidence for the **neutron**, about 20 years after scientists had accepted that atoms have a nucleus.

Why models change

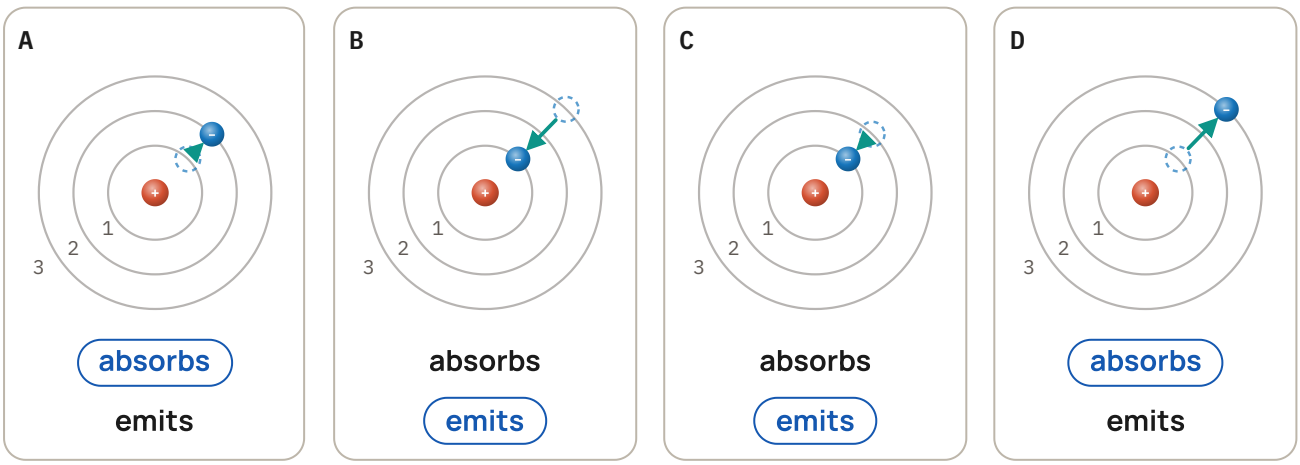
1. A model must **explain the evidence** we already have.
2. It makes **predictions** that experiments can test.
3. If new evidence does not fit, the model is **changed or replaced**.
4. Results are **published** so other scientists can check them (**peer review**) before a model is accepted.



PRACTISE

Try it

1 In each diagram an electron moves between energy levels. Does the atom **absorb** or **emit** EM radiation? Circle one for each.



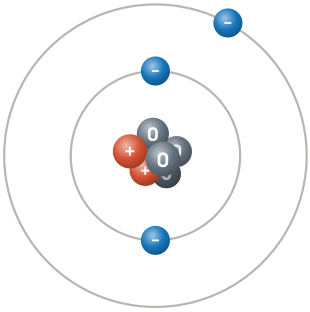
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2 Complete the sentences.

When an electron absorbs electromagnetic radiation, it moves to a ___higher___ energy level, ___further from___ the nucleus. When an electron moves to a lower energy level, it ___emits___ electromagnetic radiation.

WORD BANK: higher | lower | further from | closer to | absorbs | emits

3 A lithium atom has 3 electrons: 2 in the first energy level and 1 in the second. Draw the electrons on the diagram.



4 SUPPORTED Complete the timeline.

Year	Scientist(s)	Discovery or model
1897	J. J. Thomson	The electron (then the plum pudding model, 1904)
1909–1911	Rutherford, Geiger and Marsden	Alpha scattering; the nuclear model
1913	Niels Bohr	Electrons in energy levels
1932	James Chadwick	Evidence for the neutron

5 Explain why scientists really change models.

Leo: Scientists keep changing the model of the atom because they get bored of the old one.

When new evidence (e.g. from an experiment) does not fit the old model, the model has to be changed or replaced so it explains all the evidence. The new model is checked by other scientists (peer review) before it is accepted.

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6 CHALLENGE Protons were identified in 1919, but the neutron was not found until 1932. Suggest why the neutron was harder to find.

A neutron has no charge, so it is not deflected by electric or magnetic fields and does not ionise atoms directly, so it does not show up easily in detectors.

CHECKPOINT

Quick check

Slide 25, Lessons 1–4. Answer each question, then check it as your teacher reveals the answer.

1 What is in the nucleus?

Protons and neutrons.

2 How many neutrons are in $^{37}_{17}Cl$?

37 − 17 = 20

3 Why did most alpha particles pass straight through the foil?

The atom is mostly empty space.

4 Why did a very few alpha particles bounce back?

The nucleus is tiny, dense and positive.

5 An electron absorbs EM radiation. What happens to it?

It moves to a higher energy level (further from the nucleus).

6 Who gave the evidence for the neutron?

James Chadwick.

AQA-STYLE

Exam corner

Write in full sentences. Show every step of a calculation.

Get the marks: Use the key words: absorb/emit, higher/lower energy level, further from/closer to the nucleus.

7 Describe how the arrangement of the electrons in an atom may change when the atom absorbs electromagnetic radiation. [2 marks]

An electron moves to a higher energy level (1), further from the nucleus (1).

8 Explain why Bohr's model of the atom was accepted by other scientists. [2 marks]

His theoretical calculations agreed with experimental observations (1); it could be tested and checked by other scientists / explained evidence the nuclear model could not (1).

9 Give the name of the scientist who provided the evidence for neutrons. [1 mark]

James Chadwick.

REVIEW LATER

Revise it

Use this box after the lesson and before tests.

SUMMARY

- Bohr (1913): electrons orbit at specific distances, called energy levels. His calculations agreed with experiments.
- Absorb EM radiation → electron moves to a higher level, further from the nucleus.
- Emit EM radiation → electron moves to a lower level, closer to the nucleus.
- Protons were identified later (1919); James Chadwick gave evidence for neutrons (1932).
- New evidence changes models; results are published and peer reviewed.

COVER THE PAGE AND ANSWER

1 What did Bohr add to the nuclear model?

2 What happens when an electron emits EM radiation?

3 Who gave the evidence for the neutron?

4 Why are scientific results published?

I CAN…

- describe Bohr's model of the atom
- explain what happens when electrons absorb or emit EM radiation
- describe the discovery of protons and neutrons
- explain why models of the atom have changed

REFERENCE

Toolkit.

The facts and methods you have met so far. Learn these: exam questions expect you to recall them without a data sheet.

The three particles

Lesson 1

Particle	Charge	Mass	Where
Proton	+1	1	nucleus
Neutron	0	1	nucleus
Electron	−1	very small	energy levels

Reading the symbol

Lesson 1

$^{A}_{Z}X$ e.g. $^{23}_{11}Na$

- Z, atomic number = protons
- A, mass number = protons + neutrons
- neutrons = A − Z; in an atom, electrons = Z
- Atom radius ≈ 1×10^{-10} m; nucleus radius < 1/10 000 of that

Isotopes and ions

Lesson 2

- **Isotopes**: same number of protons, different numbers of neutrons.
- **Positive ion**: an atom that has lost one or more outer electrons. The nucleus is unchanged.

Alpha scattering

Lesson 3

Observation	Conclusion
most straight through	mostly empty space
some deflected	positive centre
a very few bounced back	mass in a tiny centre

Models and energy levels

Lesson 4

- 1897 electron → 1904 plum pudding → 1911 nuclear → 1913 Bohr → 1932 neutron (Chadwick)
- **Absorb** EM radiation → higher level, further out.
- **Emit** EM radiation → lower level, closer in.

REFERENCE

Key words from this booklet.

Learn the meaning and spelling. In exams, precise words earn marks: say “nucleus”, not “middle”, and “unstable nucleus”, not “unstable atom”.

Absorb: Take in energy. An electron that absorbs EM radiation moves to a higher energy level.

Alpha scattering: The experiment in which alpha particles fired at thin gold foil showed that atoms have a tiny, positive nucleus.

Bohr model: Electrons orbit the nucleus at specific distances, called energy levels (1913).

Emit: Give out. An electron that moves to a lower energy level emits EM radiation.

Energy level: A specific distance from the nucleus at which electrons are found.

Evidence: Observations and measurements used to test a model.

Model: A simplified explanation of something we cannot see directly; changed when new evidence does not fit.

Nuclear model: A tiny, dense, positive nucleus with electrons outside it (Rutherford, 1911).

Peer review: Checking of published findings by other scientists.

Plum pudding model: A ball of positive charge with negative electrons embedded in it (1904).

CHECK YOURSELF

Quick answers.

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

LESSON 3: FROM PLUM PUDDING TO THE NUCLEUS

DO NOW

1. Protons and neutrons.
2. Positive: protons are positive and neutrons have no charge.
3. 1/10 000

COVER THE PAGE

1. The electron (atoms can be divided).
2. Very thin gold foil.
3. The mass (and positive charge) is concentrated in a tiny centre, the nucleus.
4. In a tiny nucleus at the centre.

LESSON 4: ENERGY LEVELS AND THE NUCLEUS

DO NOW

1. The alpha scattering experiment.
2. Positive (+2).
3. Outside the nucleus.

COVER THE PAGE

1. Electrons orbit at specific distances (energy levels).
2. It moves to a lower energy level, closer to the nucleus.
3. James Chadwick (1932).
4. So other scientists can check them (peer review).